

SENSEAID

Smart Environment Navigation and Sensory Aid Device

Using AI and YOLOv8, these smart glasses help visually impaired users navigate safely with real-time audio and vibration guidance for obstacles and turns.

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Table of Content

- Background & Opportunity
- Our Solution: SENSEAID
  - System Overview
  - Goals
  - Competitive Adventages
  - SWOT Analysis
- Business Concept & Financial Analysis
- Design Stage
- Vision: The Endless Future

Background & Opportunity

Over **250 million people** are visually impaired globally.
In China alone, over 17 million are affected, with ~5 million blind.

Traditional tools like **canes** have limitations in detecting high or moving obstacles.
Guide Dogs: Costly, require extensive training and care

Advances in **computer vision and AI** create new opportunities for assistive technologies.

This project introduces an **innovative wearable** that leverages AI and sensors to improve navigation safety.

Our Solution: SENSEAID

(Smart Environment Navigation and Sensory Aid Device)

A smart glasses system powered by Artificial Intelligence (AI) to assist visually impaired individuals.



Equipped with a camera, vibration sensors, and YOLOv8 object detection for real-time guidance.

Provides voice and vibration feedback to guide users safely in their surroundings.

Aims to enhance independence and mobility for visually impaired people.

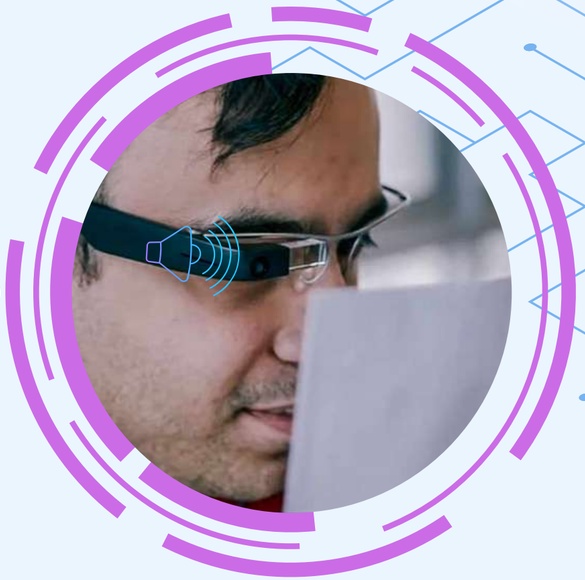
System Overview



Camera captures the environment in real-time.

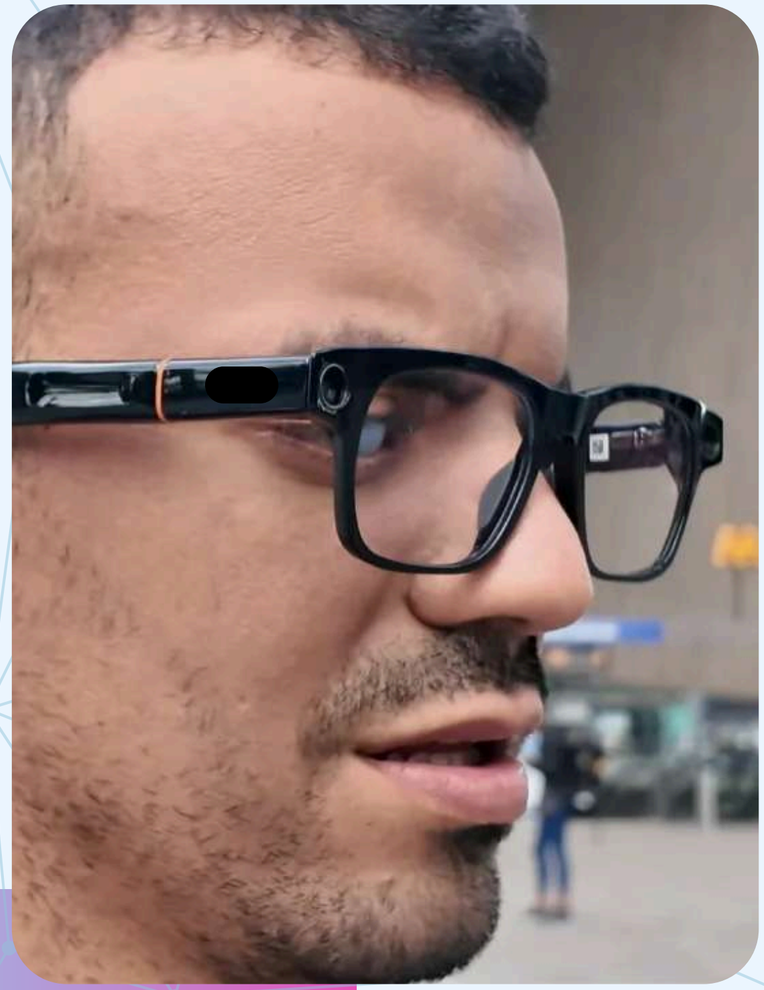


YOLOv8 processes video frames to detect obstacles and objects.



The system converts detection data into audio and vibration alerts.

Hardware: Microcontroller, camera module, haptic motor, and speaker.



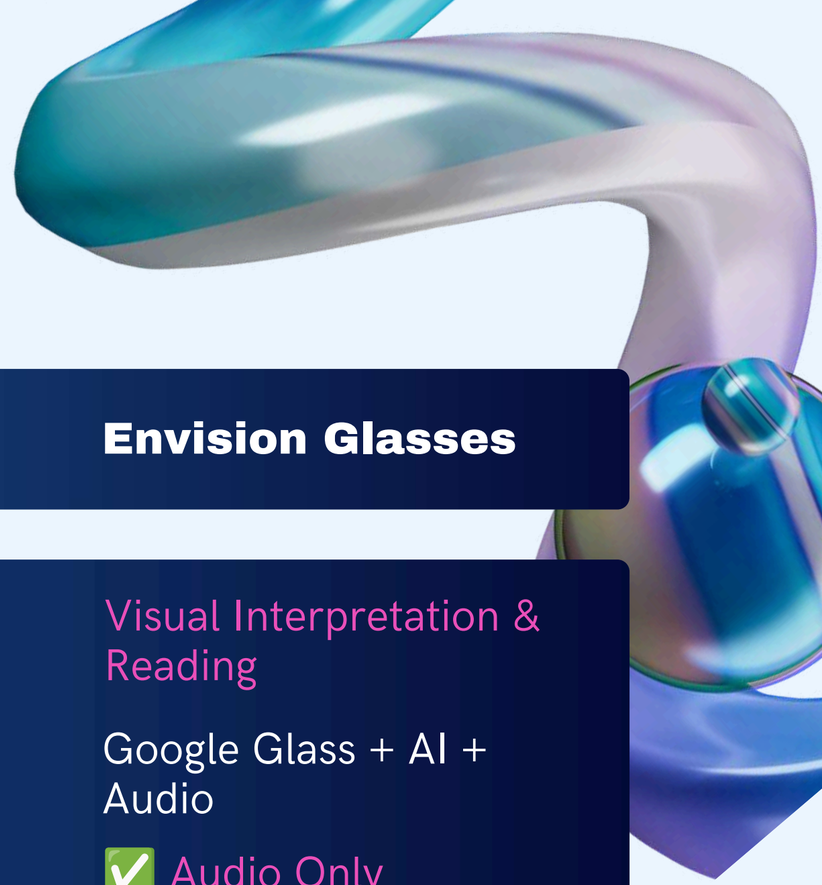
Goals

Main Goal: Enable independent and safe navigation for visually impaired individuals.

Specific Objectives:

- Detect surrounding objects in real time using YOLOv8.
- Provide distance and direction feedback via voice.
- Alert users to nearby obstacles using vibration signals.
- Develop a lightweight and user-friendly device.

Competitive Advantage



Feature	SENSEAID	OrCam MyEye	Envision Glasses
Primary Function	Mobility & Obstacle Navigation	Visual Interpretation & Reading	Visual Interpretation & Reading
Core Technology	YOLOv8 AI + Multi-Sensory Feedback	Proprietary AI + Audio	Google Glass + AI + Audio
Sensory Feedback	✅ Audio + Haptic (Vibration)	✅ Audio Only	✅ Audio Only
Obstacle Detection	Dynamic & Static (360° awareness)	Limited / Not Primary	Limited / Not Primary
Real-time Guidance	Continuous & Intuitive	On-Command / Interpretive	On-Command / Interpretive

SWOT Analysis

Strengths:

- Innovative assistive technology using AI.
- Real-time object recognition with YOLOv8.
- Human-Centered Design

Weaknesses:

- Production costs and power requirements.

Opportunities:

- Large global market for assistive technology.
- Collaboration with government and NGOs.
- Integration with Smart City Initiatives

Threats:

- Competition and hardware limitations.



Business Concept

- Product Name: senseaid Glass
- Tagline: “See the world differently.”
- Value Proposition:
  - Provides independence and confidence for users.
  - Real-time detection using AI-powered YOLOv8.
  - Lightweight, portable, and easy to use.
- Target Market: Visually impaired individuals and social organizations.
- Technology Stack: YOLOv8, ultrasonic sensors, voice and vibration output.

Market Opportunity

01	A Large and Growing Addressable Market	Targeting 17M+ visually impaired in China and 285M+ globally within the $31B+ assistive technology market, with growth driven by aging populations.
02	Favorable Market Conditions	Supported by China's "Healthy China 2030" policy and advancing AI/edge computing technologies that enable affordable solutions.
03	Clear Market Gap	Current solutions are either limited (white canes) or misdirected - competitors focus on text-reading, not navigation, while being prohibitively expensive.
04	Strategic Positioning	SENSEAID fills this gap with affordable, navigation-focused smart glasses featuring unique multi-sensory feedback for intuitive mobility assistance.

Target Customers

01 Primary End-Users (B2C - Business-to-Consumer)

Targeting 17M+ visually impaired in China and 285M+ globally within the $31B+ assistive technology market, with growth driven by aging populations.

02 Institutional & Organizational Clients (B2G/B2B - Business-to-Government/Business-to-Business)

- **Government & Public Accessibility Agencies**: Entities at the municipal and national level that implement disability inclusion policies and "smart city" accessibility programs. They would procure SENSEAID for public distribution or subsidy schemes.

- **Hospitals & Rehabilitation Centers**: Medical and therapeutic facilities that can integrate SENSEAID into their mobility training and rehabilitation services for patients.

- **Non-Governmental Organizations (NGOs)**: Organizations dedicated to supporting people with disabilities. They act as distribution partners or funders, providing devices to low-income individuals.

- **Assistive Technology Retailers & Distributors**: Companies specializing in medical and mobility devices that would sell SENSEAID through their established retail channels.

Financial Analysis

Initial Investment & Funding Breakdown



Component	Description	Cost (CNY)
Hardware Components	Camera modules, processors, audio units, vibration motors, batteries, frames	¥120,000
Software Development	YOLOv8 model training, embedded software, voice feedback systems	¥80,000
Personnel & Research	Engineers, AI developers, accessibility consultants	¥100,000
Testing & Quality Control	Simulation setups, field trials, participant testing, data collection	¥50,000
Marketing & Promotion	Branding, pilot demonstrations, exhibitions	¥30,000
Administrative & Legal	Licensing, certifications, compliance	¥20,000
Total Initial Investment		**¥400,000**

Funding Sources: A mix of Founders' Equity, Government Grants, and Private Investors/CSR Programs.

Financial Analysis

Revenue & Profitability Projection (3-Year Outlook)

Year	Phase	Price/Unit (CNY)	Units Sold	Revenue (CNY)
Year 1	Pilot Production	2,000	100	¥200,000
Year 2	Market Entry	2,000	300	¥600,000
Year 3	Scaling Up	1,800	600	¥1,080,000
Total			**1,000**	**¥1,880,000**

Profitability Target: Projected to reach breakeven by Year 2, achieving a net profit margin of 20-25% in Year 3.

Design Stage

```python
# Configuration constants for the Smart Glasses project

YOLO_MODEL_NAME = "yolov8n.pt"
SHOW_DEBUG_WINDOWS = True
VOICE_RATE_WPM = 185
ALERT_COOLDOWN = 2.0

CONF_THRESH = 0.35
IOU_THRESH = 0.45

# COCO class IDs commonly acting as obstacles
OBSTACLE_CLASSES = {0, 1, 2, 3, 5, 7}  # person,bicycle,car,motorcycle,bus,truck
TRAFFIC_LIGHT_CLASS = 9

# MiDaS depth (relative) — used only for the torch path
MIDAS_REPO = "intel-isl/MiDaS"
MIDAS_MODEL = "MiDaS_small"
DEPTH_SCALE = 0.002

# Depth engine toggle (overridden by settings.yaml)
DEPTH_MODE = "onnx"          # "onnx" or "torch"
ONNX_MODEL = "dpt_slim_384.onnx"  # file must be in ./models

# Navigation sectors & ROI
SECTORS = [
    (0.00, 0.33, "left"),
    (0.33, 0.66, "center"),
    (0.66, 1.00, "right"),
]
ROI_Y_RATIO = (0.45, 0.85)
MIN_SAFE_METERS = 1.5
```

```python
import os
os.environ["OMP_NUM_THREADS"] = "2"
os.environ["MKL_NUM_THREADS"] = "2"

import time
import cv2
import numpy as np

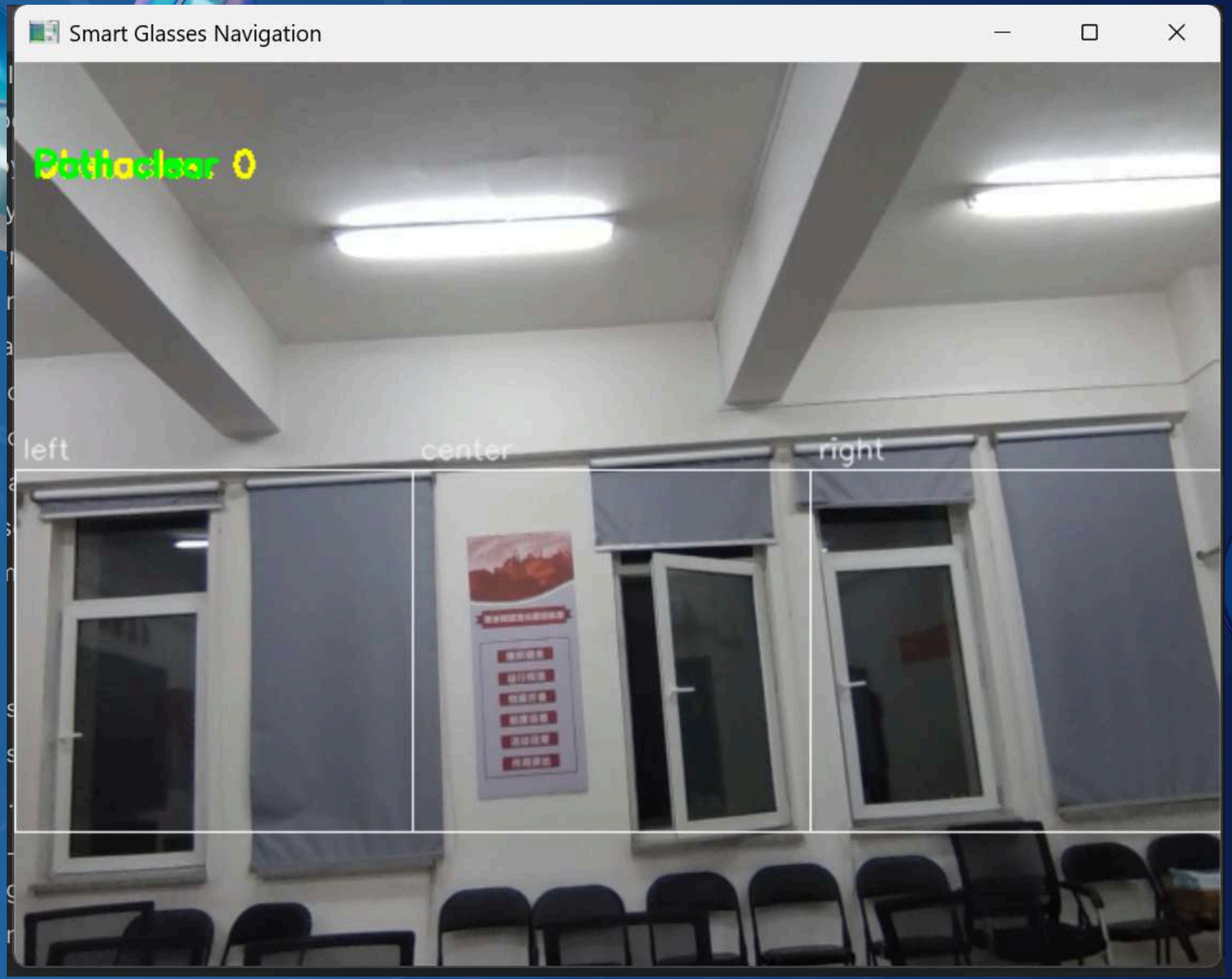
# --- load config, then apply settings.yaml overrides ---
import config as _cfg
from settings_loader import Settings

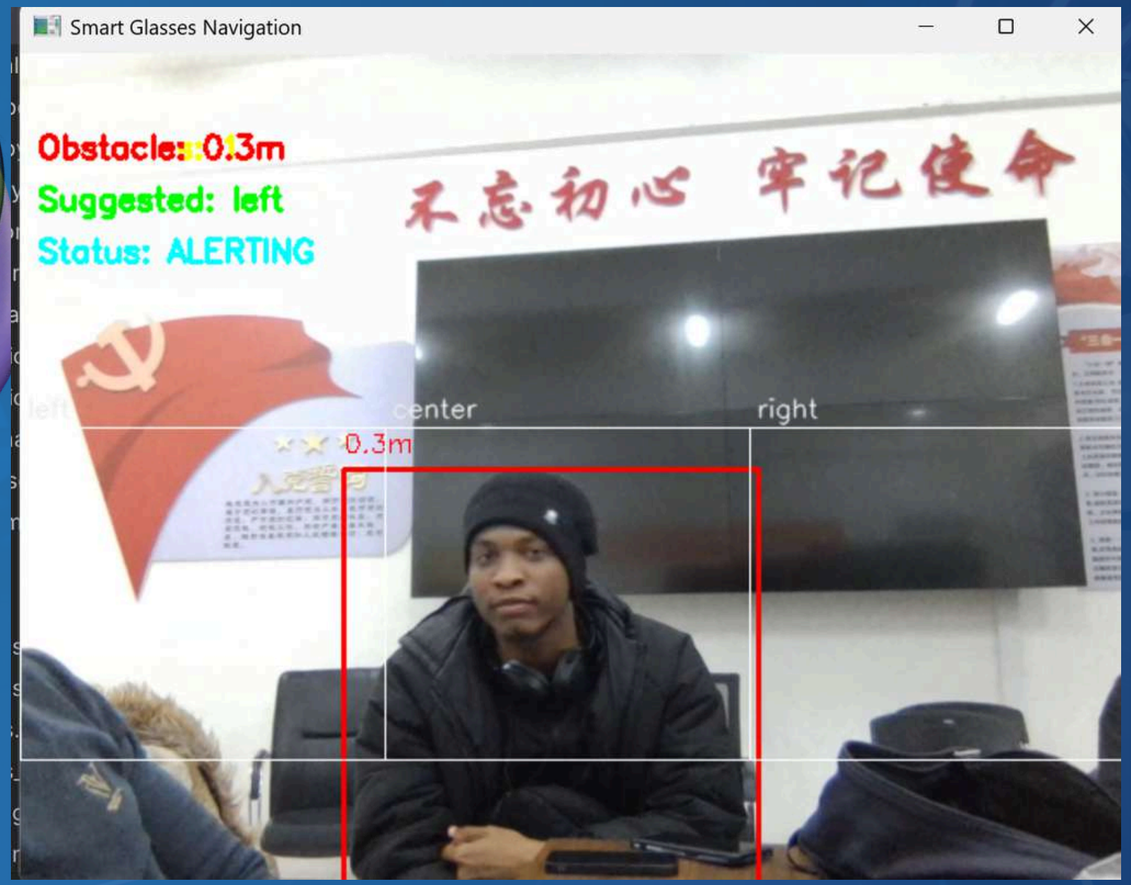
_s = Settings("settings.yaml")
_s.load()
_s.apply_to_config(_cfg)

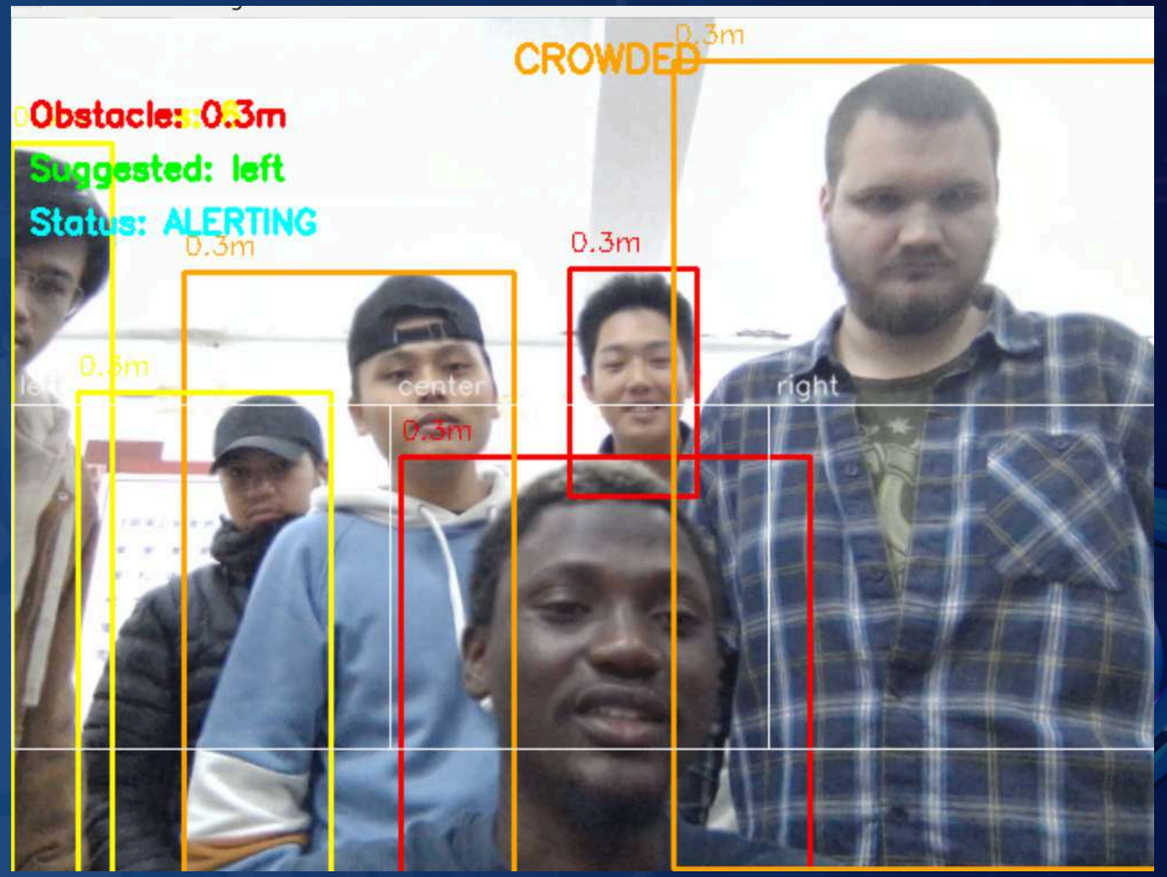
from config import (
    YOLO_MODEL_NAME, SHOW_DEBUG_WINDOWS, ALERT_COOLDOWN,
    CONF_THRESH, IOU_THRESH, OBSTACLE_CLASSES, TRAFFIC_LIGHT_CLASS,
    DEPTH_MODE, ONNX_MODEL
)

from detector import ObjectDetector
from navigation import sector_free_distance, draw_sector_boxes, best_turn_instruction
from traffic_light import classify_traffic_light_state
from tts_engine import TTSEngine

# ---- SELECT DEPTH ENGINE AUTOMATICALLY ----
if str(DEPTH_MODE).lower() == "onnx":
    try:
        from depth_onnx import DepthEstimatorONNX as DepthEstimator
        print(f"[INFO] Using ONNX Depth Engine ✅ ({ONNX_MODEL})")
    except Exception as e:
        print(f"[ERROR] ONNX failed ({e}), falling back to Torch depth.")
        from depth import DepthEstimator
else:
```


Smart Glasses Navigation
Pathclear 0
left
center
right


Smart Glasses Navigation
Obstacle: 0.3m
Suggested: left
Status: ALERTING
不忘初心 牢记使命
center
right
0.3m


CROWDED
Obstacle: 0.3m
Suggested: left
Status: ALERTING
0.3m
0.3m
0.3m
0.3m
0.3m
center
right

Vision: The Endless Future

1. Evolve into a comprehensive assistive ecosystem.
2. Integration with Smart Cities and IoT.
3. Cloud-based AI for continuous learning and personalization.
4. Global expansion to empower millions worldwide.

Our Vision: A future where independence, safety, and inclusion are no longer limited by visual impairment.

THANK YOU!